Chapter 2 Duolingo Math checklist

Tick each textbook section as you read it and each Duolingo unit as you finish it. Units are listed per lane (a Duolingo grade or topic) in the order the app presents them, so you can follow one lane top to bottom. Ratings: full = covers the section's main skill, partial = covers part of it, prereq = a warm-up from an earlier grade. The right column names the textbook sections the unit serves.

Chapter 2: Polynomial and Rational Functions (p. 125)

Grades 11 and 12 between them cover polynomial, complex, and rational functions thoroughly. Duolingo has no unit on polynomial long or synthetic division (2.3) or on nonlinear inequalities (2.7); use the book for those.

Textbook sections

☐	2.1 Quadratic Functions and Models	p. 126
☐	2.2 Polynomial Functions of Higher Degree	p. 136
☐	2.3 Polynomial and Synthetic Division	p. 150
☐	2.4 Complex Numbers	p. 159
☐	2.5 Zeros of Polynomial Functions	p. 166
☐	2.6 Rational Functions	p. 181
☐	2.7 Nonlinear Inequalities	p. 194

Book only, not on Duolingo: **2.1** Completing the square to write vertex form; Maximum and minimum word problems. **2.2** Using the Intermediate Value Theorem to locate a zero. **2.3** Polynomial long division; Synthetic division; The Remainder Theorem and the Factor Theorem. **2.4** Plotting complex numbers in the complex plane. **2.5** The Fundamental Theorem of Algebra; Descartes's Rule of Signs; Upper and lower bounds for real zeros. **2.6** Applications of rational functions. **2.7** Polynomial inequalities; Rational inequalities; The test-interval (sign chart) method.

Grade 9

☐	11. Solve one-variable inequaltiies	prereq	2.7
☐	12. Compound 'or' inequalities	prereq	2.7
☐	13. Compound 'and' inequalities	prereq	2.7
☐	56. Tables of quadratic functions	full	2.1
☐	57. Graphs of quadratic functions	full	2.1
☐	58. Equations of quadratic functions	full	2.1
☐	59. Zeros of quadratic functions	full	2.1
☐	60. Factored form	full	2.1
☐	61. Vertex	full	2.1
☐	62. Vertex form	full	2.1

Grade 11

	12. Intro to polynomial functions	full	2.2
	13. Relative maxima and minima of polynomial functions	full	2.2
	14. Zeros of polynomial functions	full	2.2, 2.5
	15. Multiplicity of polynomial functions	full	2.2
	16. End behavior of odd polynomial functions	full	2.2
	17. End behavior of even polynomial functions	full	2.2
	18. The imaginary number, i	full	2.4
	19. Square roots of negative numbers	full	2.4
	20. Patterns of exponentiating i	full	2.4
	21. Multiply complex numbers	full	2.4
	22. Complex roots	full	2.5
	23. Complex conjugates	full	2.4
	24. Intro to rational functions	full	2.6
	25. Y-intercept of rational functions	full	2.6
	26. X-intercepts of rational functions	full	2.6
	27. Vertical asymptotes of rational functions	full	2.6
	28. Rewrite rational functions	full	2.6
	29. End behavior approaching a horizontal line	full	2.6
	30. End behavior approaching y=0	full	2.6
	31. End behavior approaching slant asymptotes	full	2.6

Grade 12

	2. Polynomial function graphs	full	2.2
	3. Roots of polynomial functions	full	2.5
	4. Rational root theorem	full	2.5
	5. Polynomial function equations from graphs	full	2.2
	6. Vieta's formula for quadratic polynomials	partial	2.5
	7. Vieta's formula for cubic polynomials	partial	2.5
	8. Zeros of rational functions	full	2.6
	9. Vertical asymptotes of rational functions	full	2.6
	10. Horizontal asymptotes of rational functions	full	2.6
	11. Holes in rational functions	full	2.6
	12. Slant asymptotes of rational functions	full	2.6
	48. Intermediate value theorem	partial	2.2

Algebraic Graphing

☐	11. Intro to quadratic functions	full	2.1
☐	12. Quadratic graphs and equations	full	2.1
☐	13. Factored form of quadratics	full	2.1
☐	14. Vertex form of quadratics	full	2.1
☐	30. Polynomial function equations	full	2.2
☐	31. Polynomial function graphs	full	2.2
☐	32. Polynomial graphs and roots	full	2.2
☐	33. Rational function equations and graphs	full	2.6
☐	34. Rewrite and graph rational functions	full	2.6
☐	35. Rational function asymptotes	full	2.6
☐	36. Holes and slant asymptotes	full	2.6

Section guide

For each section: the Duolingo units that practise it (already listed above with checkboxes), then the parts only the textbook teaches. Tick each book-only item once you have worked it in the book.

2.1 Quadratic Functions and Models (p. 126)

Thoroughly covered in grade 9 and the Algebraic Graphing topic. Completing the square to reach vertex form is lighter in the app.

Duolingo, in app order:
Grade 9: 56. Tables of quadratic functions (full) · 57. Graphs of quadratic functions (full) · 58. Equations of quadratic functions (full) · 59. Zeros of quadratic functions (full) · 60. Factored form (full) · 61. Vertex (full) · 62. Vertex form (full)
Algebraic Graphing: 11. Intro to quadratic functions (full) · 12. Quadratic graphs and equations (full) · 13. Factored form of quadratics (full) · 14. Vertex form of quadratics (full)

Book only:

- [ ] Completing the square to write vertex form
- [ ] Maximum and minimum word problems

2.2 Polynomial Functions of Higher Degree (p. 136)

Grade 11 covers end behaviour, zeros, multiplicity, and relative extrema; grade 12 adds writing equations from graphs. The Leading Coefficient Test is taught as end behaviour.

Duolingo, in app order:
Grade 11: 12. Intro to polynomial functions (full) · 13. Relative maxima and minima of polynomial functions (full) · 14. Zeros of polynomial functions (full) · 15. Multiplicity of polynomial functions (full) · 16. End behavior of odd polynomial functions (full) · 17. End behavior of even polynomial functions (full)
Grade 12: 2. Polynomial function graphs (full) · 5. Polynomial function equations from graphs (full) · 48. Intermediate value theorem (partial)
Algebraic Graphing: 30. Polynomial function equations (full) · 31. Polynomial function graphs (full) · 32. Polynomial graphs and roots (full)

Book only:

- [ ] Using the Intermediate Value Theorem to locate a zero

2.3 Polynomial and Synthetic Division (p. 150)

Duolingo has no unit on polynomial long division, synthetic division, or the Remainder and Factor Theorems. Use the book; grade 12's Rational root theorem assumes this skill.

Duolingo: no unit covers this section.

Book only:

- [ ] Polynomial long division
- [ ] Synthetic division
- [ ] The Remainder Theorem and the Factor Theorem

2.4 Complex Numbers (p. 159)

Grade 11 covers the imaginary unit, arithmetic, conjugates, and powers of i. Division by a complex number is practised through conjugates; the complex plane is book only.

Duolingo, in app order:
Grade 11: 18. The imaginary number, i (full) · 19. Square roots of negative numbers (full) · 20. Patterns of exponentiating i (full) · 21. Multiply complex numbers (full) · 23. Complex conjugates (full)

Book only:

- [ ] Plotting complex numbers in the complex plane

2.5 Zeros of Polynomial Functions (p. 166)

Rational zero test, complex zeros, and conjugate pairs are covered across grades 11 and 12. Vieta's formulas go beyond the section. Descartes' Rule of Signs and bounds are book only.

Duolingo, in app order:
Grade 11: 14. Zeros of polynomial functions (prereq) · 22. Complex roots (full)
Grade 12: 3. Roots of polynomial functions (full) · 4. Rational root theorem (full) · 6. Vieta's formula for quadratic polynomials (partial) · 7. Vieta's formula for cubic polynomials (partial)

Book only:

- [ ] The Fundamental Theorem of Algebra
- [ ] Descartes's Rule of Signs
- [ ] Upper and lower bounds for real zeros

2.6 Rational Functions (p. 181)

Very well covered. Grade 11 walks through intercepts and each kind of asymptote; grade 12 adds holes and slant asymptotes explicitly.

Duolingo, in app order:
Grade 11: 24. Intro to rational functions (full) · 25. Y-intercept of rational functions (full) · 26. X-intercepts of rational functions (full) · 27. Vertical asymptotes of rational functions (full) · 28. Rewrite rational functions (full) · 29. End behavior approaching a horizontal line (full) · 30. End behavior approaching y=0 (full) · 31. End behavior approaching slant asymptotes (full)
Grade 12: 8. Zeros of rational functions (full) · 9. Vertical asymptotes of rational functions (full) · 10. Horizontal asymptotes of rational functions (full) · 11. Holes in rational functions (full) · 12. Slant asymptotes of rational functions (full)
Algebraic Graphing: 33. Rational function equations and graphs (full) · 34. Rewrite and graph rational functions (full) · 35. Rational function asymptotes (full) · 36. Holes and slant asymptotes (full)

Book only:

- [ ] Applications of rational functions

2.7 Nonlinear Inequalities (p. 194)

Duolingo has no unit on polynomial or rational inequalities or the sign-chart method. Grade 9 one-variable inequalities are the only warm-up.

Duolingo, in app order:
Grade 9: 11. Solve one-variable inequaltiies (prereq) · 12. Compound 'or' inequalities (prereq) · 13. Compound 'and' inequalities (prereq)

Book only:

- [ ] Polynomial inequalities
- [ ] Rational inequalities
- [ ] The test-interval (sign chart) method

Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Duolingo inventory: 820 units, last verified 2026-10-07 on iOS. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.